

QuantWise

An AI-Powered Decision-Support Stock Advisory Platform

ABSTRACT

The rapid rise of commission-free trading applications has made stock-market participation accessible to millions of non-expert retail investors, yet the financial knowledge required to invest wisely has not followed. New investors face information overload and often fall back on generic robo-advice or guesswork, leading to avoidable losses. This project presents **QuantWise**, an AI-powered decision-support platform that delivers personalised, risk-graded investment recommendations to retail investors without requiring financial expertise. QuantWise combines a hybrid two-stage forecasting model — a Long Short-Term Memory (LSTM) network that encodes 60-day price sequences into temporal embeddings, concatenated with hand-crafted technical indicators and passed to an XGBoost regressor — with FinBERT-based financial-news sentiment analysis and a deterministic risk-grading engine. A constrained Large Language Model (Google Gemini) then synthesises these pre-validated signals into personalised BUY / SELL / HOLD recommendations, each accompanied by a plain-language rationale and tailored to the user's risk profile; the model is restricted to validated data and never forecasts or invents figures, mitigating hallucination. The system is engineered as a .NET modular monolith with a FastAPI machine-learning pipeline, a React front end, and PostgreSQL, Redis, and RabbitMQ infrastructure. The hybrid model achieves a 30-day test RMSE of 0.0949, roughly one-third that of a standalone LSTM baseline, and was validated through 68 automated tests, load testing, and usability evaluation. By making professional-grade, explainable investment guidance instantly accessible, QuantWise helps first-time investors beat inflation safely, aligning with UN Sustainable Development Goals 9 and 10.

KEYWORDS

decision-support systems · stock-market prediction · LSTM · XGBoost · hybrid model ·
sentiment analysis · FinBERT · large language models · risk profiling · personalised
recommendations · FinTech

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